

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12385854
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Dinca; Dan-Cristian

Methods and systems for performing on-the-fly automatic calibration adjustments of X-ray inspection systems

Abstract

The specification discloses methods of adjusting calibration data in an X-ray inspection system. Calibration data is initially generated. X-ray scan images of a cargo container are then acquired. Each of the X-ray scan images are segmented into regions of interest, where the regions of interest volumetrically encompass a known material or a material corresponding to a known HS code. Using the calibration data, first data indicative of Zeff of each of the regions of interest are determined. The first data is compared with second data indicative of known Zeff corresponding to the known materials and/or HS codes. The calibration data is then adjusted to generate a second calibration data if the first and second data differ significantly. The calibration data is replaced by the second calibration data in the memory.

Inventors:	Dinca; Dan-Cristian (Chelmsford, MA)
Applicant:	Rapiscan Holdings, Inc. (Hawthorne, CA)
Family ID:	86228749
Assignee:	Rapiscan Holdings, Inc. (Hawthorne, CA)
Appl. No.:	18/149401
Filed:	January 03, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230147681 A1	May. 11, 2023

Related U.S. Application Data

us-provisional-application US 63369386 20220726

Publication Classification

Int. Cl.: G01N23/04 (20180101); G01V5/22 (20240101)

U.S. Cl.:

CPC G01N23/04 (20130101); G01V5/224 (20240101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2831123	12/1957	Daly	N/A	N/A
2971433	12/1960	Akin	N/A	N/A
2972430	12/1960	Johnson	N/A	N/A
3374355	12/1967	Parratt	N/A	N/A
3417243	12/1967	Hill	N/A	N/A
3676783	12/1971	Kinbara	N/A	N/A
3766387	12/1972	Heffan	N/A	N/A
3767850	12/1972	Mc Millian	N/A	N/A
3770955	12/1972	Tomita	N/A	N/A
3784837	12/1973	Holmstrom	N/A	N/A
3904923	12/1974	Schwartz	N/A	N/A
3961186	12/1975	Leunbach	N/A	N/A
3988586	12/1975	Stuart	N/A	N/A
4020346	12/1976	Dennis	N/A	N/A
4031401	12/1976	Jacob	N/A	N/A
4047035	12/1976	Dennhoven	N/A	N/A
4064440	12/1976	Roder	N/A	N/A
4139771	12/1978	Dennhoven	N/A	N/A
4210811	12/1979	Dennhoven	N/A	N/A
4216499	12/1979	Manfred	N/A	N/A
4242583	12/1979	Annis	N/A	N/A
4254599	12/1980	Maistre	N/A	N/A
4260898	12/1980	Annis	N/A	N/A
4315146	12/1981	Rudin	N/A	N/A
4342914	12/1981	Bjorkholm	N/A	N/A
4366382	12/1981	Kotowski	N/A	N/A
4380817	12/1982	Harding	N/A	N/A
4420182	12/1982	Kaneshiro	N/A	N/A
4430568	12/1983	Yoshida	N/A	N/A
4472822	12/1983	Swift	N/A	N/A
4503332	12/1984	Annis	N/A	N/A
4525854	12/1984	Molbert	N/A	N/A
4563707	12/1985	Kishida	N/A	N/A

4566113	12/1985	Gerhard	N/A	N/A
4599740	12/1985	Cable	N/A	N/A
4626688	12/1985	Barnes	N/A	N/A
4641330	12/1986	Herwig	N/A	N/A
4646339	12/1986	Rice	N/A	N/A
4709382	12/1986	Sones	N/A	N/A
4719153	12/1987	Akasawa	N/A	N/A
4736401	12/1987	Donges	N/A	N/A
4788704	12/1987	Donges	N/A	N/A
4799247	12/1988	Annis	N/A	N/A
4809312	12/1988	Annis	N/A	N/A
4817123	12/1988	Sones	N/A	N/A
4825454	12/1988	Annis	N/A	N/A
4853595	12/1988	Alfano	N/A	N/A
4864142	12/1988	Gomberg	N/A	N/A
4870670	12/1988	Geus	N/A	N/A
4872188	12/1988	Lauro	N/A	N/A
4884289	12/1988	Glockmann	N/A	N/A
4906973	12/1989	Karbowski	N/A	N/A
4975917	12/1989	Villa	N/A	N/A
4979202	12/1989	Siczek	N/A	N/A
4991189	12/1990	Boomgaarden	N/A	N/A
5006299	12/1990	Gozani	N/A	N/A
5014293	12/1990	Boyd	N/A	N/A
5022062	12/1990	Annis	N/A	N/A
5046846	12/1990	Ray	N/A	N/A
5065418	12/1990	Bermbach	N/A	N/A
5076993	12/1990	Sawa	N/A	N/A
5091924	12/1991	Bermbach	N/A	N/A
5098640	12/1991	Gozani	N/A	N/A
5103099	12/1991	Bourdinaud	N/A	N/A
5114662	12/1991	Gozani	N/A	N/A
5179581	12/1992	Annis	N/A	N/A
5181234	12/1992	Smith	N/A	N/A
5182764	12/1992	Peschmann	N/A	N/A
5185778	12/1992	Magram	N/A	N/A
5202932	12/1992	Cambier	N/A	N/A
5218202	12/1992	Evers	N/A	N/A
5221843	12/1992	Alvarez	N/A	N/A
5224144	12/1992	Annis	N/A	N/A
5237598	12/1992	Albert	N/A	N/A
5247561	12/1992	Kotowski	N/A	N/A
5253283	12/1992	Annis	N/A	N/A
5313511	12/1993	Annis	N/A	N/A
5319547	12/1993	Krug	N/A	N/A
5361840	12/1993	Matthews	N/A	N/A
5367552	12/1993	Peschmann	N/A	N/A
5379334	12/1994	Zimmer	N/A	N/A
5391878	12/1994	Petroff	N/A	N/A
5394454	12/1994	Harding	N/A	N/A

5428657	12/1994	Papanicolaopoulos	N/A	N/A
5430787	12/1994	Norton	N/A	N/A
5493596	12/1995	Annis	N/A	N/A
5524133	12/1995	Neale	N/A	N/A
5548123	12/1995	Perez-Mendez	N/A	N/A
5548630	12/1995	Hell	N/A	N/A
5600303	12/1996	Husseiny	N/A	N/A
5600700	12/1996	Krug	N/A	N/A
5602894	12/1996	Bardash	N/A	N/A
5606167	12/1996	Miller	N/A	N/A
5608214	12/1996	Baron	N/A	N/A
5638420	12/1996	Armistead	N/A	N/A
5642393	12/1996	Krug	N/A	N/A
5642394	12/1996	Rothschild	N/A	N/A
5666393	12/1996	Annis	N/A	N/A
5687210	12/1996	Maitrejean	N/A	N/A
5692028	12/1996	Geus	N/A	N/A
5692029	12/1996	Husseiny	N/A	N/A
5696806	12/1996	Grodzins	N/A	N/A
5751837	12/1997	Watanabe	N/A	N/A
5763886	12/1997	Schulte	N/A	N/A
5763903	12/1997	Mandai	N/A	N/A
5764683	12/1997	Swift	N/A	N/A
5768334	12/1997	Maitrejean	N/A	N/A
5783829	12/1997	Sealock	N/A	N/A
5787145	12/1997	Geus	N/A	N/A
5790685	12/1997	Sallee	N/A	N/A
5805660	12/1997	Perion	N/A	N/A
5838758	12/1997	Krug	N/A	N/A
5838759	12/1997	Armistead	N/A	N/A
5903623	12/1998	Swift	N/A	N/A
5910973	12/1998	Grodzins	N/A	N/A
5930326	12/1998	Rothschild	N/A	N/A
5940468	12/1998	Huang	N/A	N/A
5949842	12/1998	Schafer	N/A	N/A
5974111	12/1998	Krug	N/A	N/A
6018562	12/1999	Willson	N/A	N/A
6031890	12/1999	Bermbach	N/A	N/A
6058158	12/1999	Eiler	N/A	N/A
6067344	12/1999	Grodzins	N/A	N/A
6069936	12/1999	Bjorkholm	N/A	N/A
6078052	12/1999	Difilippo	N/A	N/A
6081580	12/1999	Grodzins	N/A	N/A
6094472	12/1999	Smith	N/A	N/A
6118850	12/1999	Mayo	N/A	N/A
6125165	12/1999	Warburton	N/A	N/A
6151381	12/1999	Grodzins	N/A	N/A
6188747	12/2000	Geus	N/A	N/A
6192101	12/2000	Grodzins	N/A	N/A
6192104	12/2000	Adams	N/A	N/A

6195413	12/2000	Geus	N/A	N/A
6198795	12/2000	Naumann	N/A	N/A
6212251	12/2000	Tomura	N/A	N/A
6216540	12/2000	Nelson	N/A	N/A
6218943	12/2000	Ellenbogexn	N/A	N/A
6236709	12/2000	Perry	N/A	N/A
6249567	12/2000	Rothschild	N/A	N/A
6252929	12/2000	Swift	N/A	N/A
6256369	12/2000	Lai	N/A	N/A
6278115	12/2000	Annis	N/A	N/A
6282260	12/2000	Grodzins	N/A	N/A
6292533	12/2000	Swift	N/A	N/A
6301326	12/2000	Bjorkholm	N/A	N/A
6304629	12/2000	Conway	N/A	N/A
6320933	12/2000	Grodzins	N/A	N/A
6324243	12/2000	Edic	N/A	N/A
6347132	12/2001	Annis	N/A	N/A
6356620	12/2001	Rothschild	N/A	N/A
6418194	12/2001	McPherson	N/A	N/A
6421420	12/2001	Grodzins	N/A	N/A
6424695	12/2001	Grodzins	N/A	N/A
6434219	12/2001	Rothschild	N/A	N/A
6435715	12/2001	Betz	N/A	N/A
6442233	12/2001	Grodzins	N/A	N/A
6445765	12/2001	Frank	N/A	N/A
6448564	12/2001	Johnson	N/A	N/A
6453003	12/2001	Springer	N/A	N/A
6453007	12/2001	Adams	N/A	N/A
6456684	12/2001	Mun	N/A	N/A
6459761	12/2001	Grodzins	N/A	N/A
6459764	12/2001	Chalmers	N/A	N/A
6473487	12/2001	Le	N/A	N/A
RE37899	12/2001	Grodzins	N/A	N/A
6483894	12/2001	Hartick	N/A	N/A
6507025	12/2002	Verbinski	N/A	N/A
6532276	12/2002	Hartick	N/A	N/A
6542574	12/2002	Grodzins	N/A	N/A
6542578	12/2002	Ries	N/A	N/A
6542580	12/2002	Carver	N/A	N/A
6543599	12/2002	Jasinetzky	N/A	N/A
6546072	12/2002	Chalmers	N/A	N/A
6552346	12/2002	Verbinski	N/A	N/A
6556653	12/2002	Hussein	N/A	N/A
6563903	12/2002	Kang	N/A	N/A
6567496	12/2002	Sychev	N/A	N/A
6580778	12/2002	Meder	N/A	N/A
6584170	12/2002	Aust	N/A	N/A
6590956	12/2002	Fenkart	N/A	N/A
6597760	12/2002	Beneke	N/A	N/A
6605473	12/2002	Hajduk	N/A	N/A

6606516	12/2002	Levine	N/A	N/A
6636581	12/2002	Sorenson	N/A	N/A
6637266	12/2002	Froom	N/A	N/A
6653588	12/2002	Gillard-Hickman	N/A	N/A
6658087	12/2002	Chalmers	N/A	N/A
6663280	12/2002	Doenges	N/A	N/A
6665373	12/2002	Kotowski	N/A	N/A
6665433	12/2002	Roder	N/A	N/A
6702459	12/2003	Barnes	N/A	N/A
6727506	12/2003	Mallette	N/A	N/A
6735279	12/2003	Jacobs	N/A	N/A
6763635	12/2003	Lowman	N/A	N/A
6777684	12/2003	Volkov	N/A	N/A
6785357	12/2003	Bernardi	N/A	N/A
6812426	12/2003	Kotowski	N/A	N/A
6816571	12/2003	Bijjani	N/A	N/A
6837422	12/2004	Meder	N/A	N/A
6839134	12/2004	Saito	N/A	N/A
6839403	12/2004	Kotowski	N/A	N/A
6843599	12/2004	Le	N/A	N/A
6856344	12/2004	Frantz	N/A	N/A
6876719	12/2004	Ozaki	N/A	N/A
6879657	12/2004	Hoffman	N/A	N/A
6920197	12/2004	Kang	N/A	N/A
6928137	12/2004	Bruder	N/A	N/A
6928141	12/2004	Carver	N/A	N/A
6965314	12/2004	Bohinc, Jr.	N/A	N/A
6973158	12/2004	Besson	N/A	N/A
6987833	12/2005	Du	N/A	N/A
7010094	12/2005	Grodzins	N/A	N/A
7039159	12/2005	Muenchau	N/A	N/A
7045788	12/2005	Iwatschenko-Borho	N/A	N/A
7062011	12/2005	Tybinkowski	N/A	N/A
7099434	12/2005	Adams	N/A	N/A
7103137	12/2005	Seppi	N/A	N/A
7115875	12/2005	Worstell	N/A	N/A
7116235	12/2005	Alioto	N/A	N/A
7130484	12/2005	August	N/A	N/A
RE39396	12/2005	Swift	N/A	N/A
7151447	12/2005	Willms	N/A	N/A
7158611	12/2006	Heismann	N/A	N/A
7162005	12/2006	Bjorkholm	N/A	N/A
7166844	12/2006	Gormley	N/A	N/A
7177391	12/2006	Chapin	N/A	N/A
7185206	12/2006	Goldstein	N/A	N/A
7203276	12/2006	Arsenault	N/A	N/A
7207713	12/2006	Lowman	N/A	N/A
7215737	12/2006	Li	N/A	N/A
7218704	12/2006	Adams	N/A	N/A
7238951	12/2006	Disdier	N/A	N/A

7244947	12/2006	Polichar	N/A	N/A
7260255	12/2006	Polichar	N/A	N/A
7261466	12/2006	Bhatt	N/A	N/A
7277526	12/2006	Rifkin	N/A	N/A
7286638	12/2006	Ledoux	N/A	N/A
7308076	12/2006	Studer	N/A	N/A
7319737	12/2007	Singh	N/A	N/A
7322745	12/2007	Agrawal	N/A	N/A
7333587	12/2007	De Man	N/A	N/A
7349525	12/2007	Morton	N/A	N/A
7352843	12/2007	Hu	N/A	N/A
7354197	12/2007	Bhatt	N/A	N/A
7366282	12/2007	Peschmann	N/A	N/A
7368717	12/2007	Verbinski	N/A	N/A
7369463	12/2007	Van Dullemen	N/A	N/A
7369642	12/2007	Eilbert	N/A	N/A
7372040	12/2007	Polichar	N/A	N/A
7376218	12/2007	Chapin	N/A	N/A
7379530	12/2007	Hoff	N/A	N/A
7381962	12/2007	Goldberg	N/A	N/A
7397891	12/2007	Johnson	N/A	N/A
7400701	12/2007	Cason	N/A	N/A
7406192	12/2007	Schmiegel	N/A	N/A
7420174	12/2007	Kurita	N/A	N/A
7440543	12/2007	Morton	N/A	N/A
7440544	12/2007	Scheinman	N/A	N/A
7453987	12/2007	Richardson	N/A	N/A
7483511	12/2008	Bendahan	N/A	N/A
7486768	12/2008	Allman	N/A	N/A
7490984	12/2008	Bhatt	N/A	N/A
7492682	12/2008	Osakabe	N/A	N/A
7492862	12/2008	Bendahan	N/A	N/A
7492934	12/2008	Mundy	N/A	N/A
7505556	12/2008	Chalmers	N/A	N/A
7505557	12/2008	Modica	N/A	N/A
7505562	12/2008	Dinca	N/A	N/A
7505563	12/2008	Morton	N/A	N/A
7508910	12/2008	Safai	N/A	N/A
7510324	12/2008	Bhatt	N/A	N/A
7512215	12/2008	Morton	N/A	N/A
7517149	12/2008	Agrawal	N/A	N/A
7519148	12/2008	Kotowski	N/A	N/A
7522696	12/2008	Imai	N/A	N/A
7525101	12/2008	Grodzins	N/A	N/A
7526064	12/2008	Akery	N/A	N/A
7538325	12/2008	Mishin	N/A	N/A
7547888	12/2008	Cooke	N/A	N/A
7551714	12/2008	Rothschild	N/A	N/A
7551715	12/2008	Rothschild	N/A	N/A
7551718	12/2008	Rothschild	N/A	N/A

7555099	12/2008	Rothschild	N/A	N/A
7564939	12/2008	Morton	N/A	N/A
7593506	12/2008	Cason	N/A	N/A
7593510	12/2008	Rothschild	N/A	N/A
7609807	12/2008	Leue	N/A	N/A
7636418	12/2008	Anwar	N/A	N/A
7649976	12/2009	Georgeson	N/A	N/A
7653545	12/2009	Starkie	N/A	N/A
7660388	12/2009	Gray	N/A	N/A
7664230	12/2009	Morton	N/A	N/A
7671784	12/2009	Steinway	N/A	N/A
7684538	12/2009	Morton	N/A	N/A
7693261	12/2009	Robinson	N/A	N/A
7720195	12/2009	Allman	N/A	N/A
7724868	12/2009	Morton	N/A	N/A
7724869	12/2009	Wang	N/A	N/A
7738687	12/2009	Tortora	N/A	N/A
7741612	12/2009	Clothier	N/A	N/A
7742568	12/2009	Smith	N/A	N/A
7750294	12/2009	Bright	N/A	N/A
7760103	12/2009	Frank	N/A	N/A
7783003	12/2009	Clayton	N/A	N/A
7783004	12/2009	Kotowski	N/A	N/A
7800073	12/2009	Clothier	N/A	N/A
7809103	12/2009	Du	N/A	N/A
7809104	12/2009	Foland	N/A	N/A
7809109	12/2009	Mastronardi	N/A	N/A
7817776	12/2009	Agrawal	N/A	N/A
7831012	12/2009	Foland	N/A	N/A
7844027	12/2009	Harding	N/A	N/A
7844028	12/2009	Korsunsky	N/A	N/A
7860213	12/2009	Akery	N/A	N/A
7864920	12/2010	Rothschild	N/A	N/A
7873201	12/2010	Eilbert	N/A	N/A
7876879	12/2010	Morton	N/A	N/A
7876880	12/2010	Kotowski	N/A	N/A
7903789	12/2010	Morton	N/A	N/A
7924979	12/2010	Rothschild	N/A	N/A
7928400	12/2010	Diawara	N/A	N/A
7929663	12/2010	Morton	N/A	N/A
7949101	12/2010	Morton	N/A	N/A
7952079	12/2010	Neustadter	N/A	N/A
7957506	12/2010	Smith	N/A	N/A
7963695	12/2010	Kotowski	N/A	N/A
7965816	12/2010	Kravis	N/A	N/A
7995705	12/2010	Allman	N/A	N/A
7995707	12/2010	Rothschild	N/A	N/A
8000436	12/2010	Seppi	N/A	N/A
8031903	12/2010	Paresi	N/A	N/A
8039812	12/2010	Crocker	N/A	N/A

8059781	12/2010	Agrawal	N/A	N/A
8073099	12/2010	Niu	N/A	N/A
8085897	12/2010	Morton	N/A	N/A
8094784	12/2011	Morton	N/A	N/A
8113071	12/2011	Sagi-Dolev	N/A	N/A
8116428	12/2011	Gudmundson	N/A	N/A
8135110	12/2011	Morton	N/A	N/A
8173970	12/2011	Inbar	N/A	N/A
8179597	12/2011	Namba	N/A	N/A
8194822	12/2011	Rothschild	N/A	N/A
8204173	12/2011	Betcke	N/A	N/A
8223919	12/2011	Morton	N/A	N/A
8233586	12/2011	Boas	N/A	N/A
8243876	12/2011	Morton	N/A	N/A
8263938	12/2011	Bjorkholm	N/A	N/A
8275091	12/2011	Morton	N/A	N/A
8275092	12/2011	Zhang	N/A	N/A
8311313	12/2011	Gamble	N/A	N/A
8320523	12/2011	Zhang	N/A	N/A
8325871	12/2011	Grodzins	N/A	N/A
8331535	12/2011	Morton	N/A	N/A
8345819	12/2012	Mastronardi	N/A	N/A
8389941	12/2012	Bendahan	N/A	N/A
8389942	12/2012	Morton	N/A	N/A
8401270	12/2012	Eilbert	N/A	N/A
8439565	12/2012	Mastronardi	N/A	N/A
8442186	12/2012	Rothschild	N/A	N/A
8451974	12/2012	Morton	N/A	N/A
8457274	12/2012	Arodzero	N/A	N/A
8502699	12/2012	Zerwekh	N/A	N/A
8503605	12/2012	Morton	N/A	N/A
8503606	12/2012	Rothschild	N/A	N/A
8532823	12/2012	Mcelroy	N/A	N/A
8552722	12/2012	Lionheart	N/A	N/A
8559592	12/2012	Betcke	N/A	N/A
8625735	12/2013	Morton	N/A	N/A
8633823	12/2013	Armistead, Jr.	N/A	N/A
8690427	12/2013	Mastronardi	N/A	N/A
8750452	12/2013	Kaval	N/A	N/A
8766764	12/2013	Rayner	N/A	N/A
8774362	12/2013	Hughes	N/A	N/A
8804899	12/2013	Morton	N/A	N/A
8817098	12/2013	Millar	N/A	N/A
8824632	12/2013	Mastronardi	N/A	N/A
8824637	12/2013	Morton	N/A	N/A
8831176	12/2013	Morton	N/A	N/A
8837669	12/2013	Morton	N/A	N/A
8842808	12/2013	Rothschild	N/A	N/A
8861684	12/2013	Al-Kofahi	N/A	N/A
8884236	12/2013	Rothschild	N/A	N/A

8885794	12/2013	Morton	N/A	N/A
8903045	12/2013	Schubert	N/A	N/A
8903046	12/2013	Morton	N/A	N/A
8908831	12/2013	Bendahan	N/A	N/A
8923481	12/2013	Schubert	N/A	N/A
8929509	12/2014	Morton	N/A	N/A
8958526	12/2014	Morton	N/A	N/A
8971484	12/2014	Beckmann	N/A	N/A
8971487	12/2014	Mastronardi	N/A	N/A
8993970	12/2014	Morton	N/A	N/A
9001973	12/2014	Morton	N/A	N/A
9014339	12/2014	Grodzins	N/A	N/A
9020095	12/2014	Morton	N/A	N/A
9020103	12/2014	Grodzins	N/A	N/A
9046465	12/2014	Thompson	N/A	N/A
9048061	12/2014	Morton	N/A	N/A
9052271	12/2014	Grodzins	N/A	N/A
9057679	12/2014	Morton	N/A	N/A
9086497	12/2014	Bendahan	N/A	N/A
9093187	12/2014	Johnson	N/A	N/A
9093245	12/2014	Morton	N/A	N/A
9099279	12/2014	Rommel	N/A	N/A
9111331	12/2014	Parikh	N/A	N/A
9113839	12/2014	Morton	N/A	N/A
9117564	12/2014	Rommel	N/A	N/A
9146201	12/2014	Schubert	N/A	N/A
9158030	12/2014	Morton	N/A	N/A
9183647	12/2014	Morton	N/A	N/A
9189846	12/2014	Axel	N/A	N/A
9208988	12/2014	Morton	N/A	N/A
9218933	12/2014	Langeveld	N/A	N/A
9223050	12/2014	Kaval	N/A	N/A
9257208	12/2015	Rommel	N/A	N/A
9263225	12/2015	Morton	N/A	N/A
9274065	12/2015	Morton	N/A	N/A
9282258	12/2015	Kuznetsov	N/A	N/A
9285488	12/2015	Arodzero	N/A	N/A
9291582	12/2015	Grodzins	N/A	N/A
9311277	12/2015	Rinkel	N/A	N/A
9417060	12/2015	Schubert	N/A	N/A
9420677	12/2015	Morton	N/A	N/A
9442082	12/2015	Morton	N/A	N/A
9442213	12/2015	Bendahan	N/A	N/A
9465135	12/2015	Morton	N/A	N/A
9466456	12/2015	Rommel	N/A	N/A
9535019	12/2016	Rothschild	N/A	N/A
9541510	12/2016	Arodzero	N/A	N/A
9562866	12/2016	Morton	N/A	N/A
9562986	12/2016	Peyton	N/A	N/A
9576766	12/2016	Morton	N/A	N/A

9606259	12/2016	Morton	N/A	N/A
9618648	12/2016	Morton	N/A	N/A
9632205	12/2016	Morton	N/A	N/A
9632206	12/2016	Parikh	N/A	N/A
9638646	12/2016	Morton	N/A	N/A
9658343	12/2016	Arodzero	N/A	N/A
9675306	12/2016	Morton	N/A	N/A
9714920	12/2016	Lionheart	N/A	N/A
9726619	12/2016	Thompson	N/A	N/A
9747705	12/2016	Morton	N/A	N/A
9772426	12/2016	Armistead, Jr.	N/A	N/A
9823201	12/2016	Morton	N/A	N/A
9823383	12/2016	Hanley	N/A	N/A
9864076	12/2017	Stein	N/A	N/A
9867271	12/2017	Saverskiy	N/A	N/A
9891314	12/2017	Morton	N/A	N/A
9989508	12/2017	Awad	N/A	N/A
9996890	12/2017	Cinnamon	N/A	N/A
10021350	12/2017	Rayner	N/A	N/A
10048393	12/2017	Stein	N/A	N/A
10061041	12/2017	Rowland	N/A	N/A
10089956	12/2017	Awad	N/A	N/A
10107783	12/2017	Lionheart	N/A	N/A
10175381	12/2018	Morton	N/A	N/A
10210631	12/2018	Cinnamon	N/A	N/A
10228487	12/2018	Mastronardi	N/A	N/A
10254436	12/2018	Awad	N/A	N/A
10295483	12/2018	Morton	N/A	N/A
10302807	12/2018	Yu	N/A	N/A
10345479	12/2018	Langeveld	N/A	N/A
10353109	12/2018	Hanley	N/A	N/A
10366293	12/2018	Faviero	N/A	N/A
10368428	12/2018	Saverskiy	N/A	N/A
10386532	12/2018	Morton	N/A	N/A
10408967	12/2018	Morton	N/A	N/A
10422919	12/2018	Parikh	N/A	N/A
10452959	12/2018	Gautam	N/A	N/A
10453223	12/2018	Cinnamon	N/A	N/A
10504261	12/2018	Cinnamon	N/A	N/A
10509142	12/2018	Parikh	N/A	N/A
10510319	12/2018	Awad	N/A	N/A
10520612	12/2018	Stein	N/A	N/A
10527742	12/2019	Stein	N/A	N/A
10572963	12/2019	Cinnamon	N/A	N/A
10585206	12/2019	Bendahan	N/A	N/A
10593099	12/2019	Sudarsky	N/A	N/A
10598812	12/2019	Franco	N/A	N/A
10650783	12/2019	Awad	N/A	N/A
10698128	12/2019	Morton	N/A	N/A
10706335	12/2019	Gautam	N/A	N/A

10768338	12/2019	Yu	N/A	N/A
10770195	12/2019	Rothschild	N/A	N/A
10782440	12/2019	Hanley	N/A	N/A
10795047	12/2019	St-Aubin	N/A	N/A
10795048	12/2019	St-Aubin	N/A	N/A
10795049	12/2019	St-Aubin	N/A	N/A
10809414	12/2019	St-Aubin	N/A	N/A
10830920	12/2019	Parikh	N/A	N/A
10901114	12/2020	St-Aubin	N/A	N/A
10942291	12/2020	Morton	N/A	N/A
10976271	12/2020	Morton	N/A	N/A
11010605	12/2020	Nord	N/A	N/A
11099294	12/2020	Parikh	N/A	N/A
11119245	12/2020	Morton	N/A	N/A
11163076	12/2020	Stein	N/A	N/A
11193898	12/2020	Schubert	N/A	N/A
11263499	12/2021	Gautam	N/A	N/A
11276213	12/2021	Cinnamon	N/A	N/A
11280898	12/2021	Morton	N/A	N/A
11287391	12/2021	Yu	N/A	N/A
11307325	12/2021	Morton	N/A	N/A
11423592	12/2021	Cinnamon	N/A	N/A
11448777	12/2021	Stein	N/A	N/A
11561321	12/2022	Morton	N/A	N/A
11594001	12/2022	Sivakumar	N/A	N/A
11790575	12/2022	Cinnamon	N/A	N/A
11822041	12/2022	Morton	N/A	N/A
11852775	12/2022	Morton	N/A	N/A
11885752	12/2023	St-Aubin	N/A	N/A
11914085	12/2023	Ritter	N/A	N/A
11977037	12/2023	Desjeans-Gauthier	N/A	N/A
12019035	12/2023	Archambault	N/A	N/A
12056840	12/2023	Sivakumar	N/A	N/A
12169264	12/2023	Morton	N/A	N/A
12174334	12/2023	Parikh	N/A	N/A
12181422	12/2023	Manalad	N/A	N/A
12181619	12/2023	Stein	N/A	N/A
12235226	12/2024	Morton	N/A	N/A
2001/0016028	12/2000	Adams	N/A	N/A
2001/0022830	12/2000	Sommer	N/A	N/A
2001/0053202	12/2000	Mazess	N/A	N/A
2002/0063783	12/2001	Kurase	N/A	N/A
2003/0004792	12/2002	Townzen	N/A	N/A
2003/0068557	12/2002	Kumashiro	N/A	N/A
2003/0085348	12/2002	Megerle	N/A	N/A
2003/0095626	12/2002	Anderton	N/A	N/A
2003/0118246	12/2002	August	N/A	N/A
2003/0204361	12/2002	Townsend	N/A	N/A
2004/0016867	12/2003	Milshtein	N/A	N/A
2004/0017313	12/2003	Menache	N/A	N/A

2004/0017888	12/2003	Seppi	N/A	N/A
2004/0086078	12/2003	Adams	N/A	N/A
2004/0088584	12/2003	Shachar	N/A	N/A
2004/0120454	12/2003	Ellenbogen	N/A	N/A
2004/0125914	12/2003	Kang	N/A	N/A
2004/0141584	12/2003	Bernardi	N/A	N/A
2004/0178339	12/2003	Gentile	N/A	N/A
2004/0199785	12/2003	Pederson	N/A	N/A
2004/0258198	12/2003	Carver	N/A	N/A
2004/0267114	12/2003	Mundy	N/A	N/A
2005/0023479	12/2004	Grodzins	N/A	N/A
2005/0031075	12/2004	Hopkins	N/A	N/A
2005/0105680	12/2004	Nabors	N/A	N/A
2005/0111610	12/2004	De Man	N/A	N/A
2005/0117700	12/2004	Peschmann	N/A	N/A
2005/0135668	12/2004	Polichar	N/A	N/A
2005/0156734	12/2004	Zerwekh	N/A	N/A
2005/0157842	12/2004	Agrawal	N/A	N/A
2005/0161611	12/2004	Disdier	N/A	N/A
2005/0169421	12/2004	Muenchau	N/A	N/A
2005/0208290	12/2004	Patel	N/A	N/A
2005/0226383	12/2004	Rifkin	N/A	N/A
2005/0258371	12/2004	Stein	N/A	N/A
2005/0275545	12/2004	Alioto	N/A	N/A
2006/0011848	12/2005	Rushbrooke	257/E27.14	G01N 23/083
2006/0027751	12/2005	Kurita	N/A	N/A
2006/0176998	12/2005	Korsunsky	N/A	N/A
2006/0249685	12/2005	Tanaka	N/A	N/A
2006/0257005	12/2005	Bergeron	N/A	N/A
2006/0262902	12/2005	Wattenburg	N/A	N/A
2006/0284094	12/2005	Inbar	N/A	N/A
2007/0007455	12/2006	Juni	N/A	N/A
2007/0009088	12/2006	Edic	N/A	N/A
2007/0061150	12/2006	Sawano	N/A	N/A
2007/0085010	12/2006	Letant	N/A	N/A
2007/0098142	12/2006	Rothschild	N/A	N/A
2007/0110215	12/2006	Hu	N/A	N/A
2007/0140423	12/2006	Foland	N/A	N/A
2007/0147585	12/2006	Eilbert	N/A	N/A
2007/0159400	12/2006	DeJean	N/A	N/A
2007/0170375	12/2006	Tang	N/A	N/A
2007/0172129	12/2006	Tortora	N/A	N/A
2007/0183568	12/2006	Kang	N/A	N/A
2007/0189454	12/2006	Georgeson	N/A	N/A
2007/0210255	12/2006	Bjorkholm	N/A	N/A
2007/0228284	12/2006	Polichar	N/A	N/A
2007/0235655	12/2006	Rhiger	N/A	N/A
2007/0237294	12/2006	Hoff	N/A	N/A
2007/0269005	12/2006	Chalmers	N/A	N/A

2007/0272874	12/2006	Grodzins	N/A	N/A
2007/0280416	12/2006	Bendahan	N/A	N/A
2007/0280502	12/2006	Paresi	N/A	N/A
2007/0286337	12/2006	Wang	N/A	N/A
2008/0037707	12/2007	Rothschild	N/A	N/A
2008/0043913	12/2007	Annis	N/A	N/A
2008/0044801	12/2007	Modica	N/A	N/A
2008/0048872	12/2007	Frank	N/A	N/A
2008/0054893	12/2007	Humphreys	N/A	N/A
2008/0084963	12/2007	Clayton	N/A	N/A
2008/0128624	12/2007	Cooke	N/A	N/A
2008/0152081	12/2007	Cason	N/A	N/A
2008/0170655	12/2007	Bendahan	N/A	N/A
2008/0170670	12/2007	Bhatt	N/A	N/A
2008/0198970	12/2007	Kirshner	N/A	N/A
2008/0205594	12/2007	Bjorkholm	N/A	N/A
2008/0212742	12/2007	Hughes	N/A	N/A
2008/0230709	12/2007	Tkaczyk	N/A	N/A
2008/0260097	12/2007	Anwar	N/A	N/A
2008/0283761	12/2007	Robinson	N/A	N/A
2008/0298546	12/2007	Bueno	N/A	N/A
2008/0304622	12/2007	Morton	N/A	N/A
2009/0003514	12/2008	Edic	N/A	N/A
2009/0010386	12/2008	Peschmann	N/A	N/A
2009/0034790	12/2008	Song	N/A	N/A
2009/0067575	12/2008	Seppi	N/A	N/A
2009/0086314	12/2008	Namba	N/A	N/A
2009/0086907	12/2008	Smith	N/A	N/A
2009/0103686	12/2008	Rothschild	N/A	N/A
2009/0116617	12/2008	Mastronardi	N/A	N/A
2009/0127459	12/2008	Neustadter	N/A	N/A
2009/0140158	12/2008	Clothier	N/A	N/A
2009/0175411	12/2008	Gudmundson	N/A	N/A
2009/0200373	12/2008	Landwirth	N/A	N/A
2009/0200480	12/2008	Clothier	N/A	N/A
2009/0213989	12/2008	Harding	N/A	N/A
2009/0221881	12/2008	Qian	N/A	N/A
2009/0238336	12/2008	Akery	N/A	N/A
2009/0257555	12/2008	Chalmers	N/A	N/A
2009/0274277	12/2008	Morton	N/A	N/A
2009/0278683	12/2008	Carter	N/A	N/A
2009/0285353	12/2008	Ellenbogen	N/A	N/A
2009/0316855	12/2008	Morton	N/A	N/A
2010/0002834	12/2009	Gudmundson	N/A	N/A
2010/0020937	12/2009	Hautmann	N/A	N/A
2010/0034353	12/2009	Kravis	N/A	N/A
2010/0061509	12/2009	Raymond	N/A	N/A
2010/0065746	12/2009	Grazioso	N/A	N/A
2010/0086185	12/2009	Weiss	N/A	N/A
2010/0098218	12/2009	Vermilyea	N/A	N/A

2010/0161504	12/2009	Casey	N/A	N/A
2010/0166142	12/2009	Du	N/A	N/A
2010/0172476	12/2009	Morton	N/A	N/A
2010/0177868	12/2009	Smith	N/A	N/A
2010/0223016	12/2009	Gibson	N/A	N/A
2010/0284509	12/2009	Oreper	N/A	N/A
2010/0295689	12/2009	Armistead, Jr.	N/A	N/A
2011/0019797	12/2010	Morton	N/A	N/A
2011/0019799	12/2010	Shedlock	N/A	N/A
2011/0031405	12/2010	Kulik	N/A	N/A
2011/0038453	12/2010	Morton	N/A	N/A
2011/0064192	12/2010	Morton	N/A	N/A
2011/0075808	12/2010	Rothschild	N/A	N/A
2011/0096906	12/2010	Langeveld	N/A	N/A
2011/0102235	12/2010	Abdillah	N/A	N/A
2011/0127426	12/2010	Akery	N/A	N/A
2011/0135060	12/2010	Morton	N/A	N/A
2011/0172972	12/2010	Gudmundson	N/A	N/A
2011/0204243	12/2010	Bendahan	N/A	N/A
2011/0206179	12/2010	Bendahan	N/A	N/A
2011/0206240	12/2010	Hong	N/A	N/A
2011/0222733	12/2010	Smith	N/A	N/A
2011/0235777	12/2010	Gozani	N/A	N/A
2011/0273320	12/2010	Nogueira-Nine	N/A	N/A
2011/0305318	12/2010	Robinson	N/A	N/A
2012/0069963	12/2011	Song	N/A	N/A
2012/0081386	12/2011	Wiemker	N/A	N/A
2012/0093367	12/2011	Gudmundson	N/A	N/A
2012/0104276	12/2011	Miller	N/A	N/A
2012/0134473	12/2011	Morton	N/A	N/A
2012/0201354	12/2011	Kimura	N/A	N/A
2012/0219116	12/2011	Thompson	N/A	N/A
2012/0294423	12/2011	Cheung	N/A	N/A
2012/0313555	12/2011	Ho	N/A	N/A
2013/0156151	12/2012	Sugaya	N/A	N/A
2013/0170611	12/2012	Beckmann	N/A	N/A
2013/0202089	12/2012	Schubert	N/A	N/A
2013/0208857	12/2012	Arodzero	N/A	N/A
2013/0230139	12/2012	Morton	N/A	N/A
2013/0251098	12/2012	Morton	N/A	N/A
2013/0256520	12/2012	Korkin	N/A	N/A
2013/0264483	12/2012	Abenaim	N/A	N/A
2013/0278631	12/2012	Border	N/A	N/A
2013/0336447	12/2012	Morton	N/A	N/A
2014/0023181	12/2013	Noshi	N/A	N/A
2014/0028457	12/2013	Reinholdt	N/A	N/A
2014/0029725	12/2013	Ueda	N/A	N/A
2014/0063239	12/2013	Furness	N/A	N/A
2014/0211916	12/2013	Morton	N/A	N/A
2014/0270086	12/2013	Krasnykh	N/A	N/A

2014/0294147	12/2013	Chen	N/A	N/A
2014/0342631	12/2013	Morton	N/A	N/A
2015/0104089	12/2014	Litvin	N/A	N/A
2015/0186732	12/2014	Perron	N/A	N/A
2015/0325010	12/2014	Bedford	N/A	N/A
2015/0357148	12/2014	Morton	N/A	N/A
2016/0025888	12/2015	Peschmann	N/A	N/A
2016/0048984	12/2015	Frigo	N/A	N/A
2016/0055650	12/2015	Park	N/A	N/A
2016/0259412	12/2015	Flint	N/A	N/A
2016/0343533	12/2015	Morton	N/A	N/A
2017/0071559	12/2016	Kappler	N/A	N/A
2017/0161922	12/2016	Morton	N/A	N/A
2017/0215814	12/2016	Cales	N/A	N/A
2017/0227659	12/2016	Stein	N/A	N/A
2017/0358380	12/2016	Rothschild	N/A	N/A
2018/0038988	12/2017	Morton	N/A	N/A
2018/0078233	12/2017	Jin	N/A	N/A
2018/0128754	12/2017	Thompson	N/A	N/A
2018/0294066	12/2017	Rothschild	N/A	N/A
2018/0333109	12/2017	Zamenhof	N/A	N/A
2018/0368248	12/2017	Saverskiy	N/A	N/A
2019/0178821	12/2018	Morton	N/A	N/A
2019/0212279	12/2018	St-Aubin	N/A	N/A
2019/0323946	12/2018	Myers	N/A	N/A
2019/0346382	12/2018	Rothschild	N/A	N/A
2019/0346586	12/2018	Scoullar	N/A	N/A
2019/0369952	12/2018	Little	N/A	N/A
2020/0042286	12/2019	Bui	N/A	N/A
2020/0337658	12/2019	Sjöstrand	N/A	N/A
2020/0340918	12/2019	Shimizu	N/A	G01N 21/314

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1301371	12/1991	CA	N/A
2163884	12/1993	CA	N/A
2744690	12/2008	CA	N/A
2624658	12/2008	CA	N/A
2624663	12/2008	CA	N/A
2636306	12/2008	CA	N/A
2697525	12/2009	CA	N/A
2690163	12/2010	CA	N/A
1745296	12/2005	CN	N/A
102175698	12/2010	CN	N/A
103327901	12/2012	CN	N/A
104165896	12/2016	CN	N/A
107209282	12/2016	CN	N/A
0077018	12/1982	EP	N/A

0176314	12/1985	EP	N/A
0261984	12/1987	EP	N/A
0287707	12/1987	EP	N/A
0864884	12/1997	EP	N/A
0919186	12/1998	EP	N/A
0672332	12/1999	EP	N/A
1135700	12/2000	EP	N/A
1254384	12/2001	EP	N/A
1413898	12/2003	EP	N/A
1526392	12/2004	EP	N/A
1733213	12/2005	EP	N/A
1739460	12/2006	EP	N/A
1907831	12/2007	EP	N/A
2049888	12/2008	EP	N/A
2054741	12/2008	EP	N/A
2147297	12/2009	EP	N/A
3037401	12/2015	FR	N/A
516517	12/1939	GB	N/A
2023814	12/1979	GB	N/A
2084829	12/1981	GB	N/A
2150526	12/1984	GB	N/A
2255634	12/1991	GB	N/A
2258321	12/1992	GB	N/A
2277013	12/1993	GB	N/A
2424065	12/2005	GB	N/A
2438317	12/2006	GB	N/A
10017382	12/2009	GB	N/A
2002357692	12/2001	JP	N/A
201351156	12/2012	JP	N/A
2276352	12/2005	RU	N/A
9002415	12/1989	WO	N/A
1990011507	12/1989	WO	N/A
9718462	12/1996	WO	N/A
9802763	12/1997	WO	N/A
1998002763	12/1997	WO	N/A
1998003889	12/1997	WO	N/A
1998020366	12/1997	WO	N/A
1998055851	12/1997	WO	N/A
9939189	12/1998	WO	N/A
2004010127	12/2003	WO	N/A
2004097889	12/2003	WO	N/A
2004109331	12/2003	WO	N/A
2005050405	12/2004	WO	N/A
2005079437	12/2004	WO	N/A
2005098400	12/2004	WO	N/A
2005121756	12/2004	WO	N/A
2006036076	12/2005	WO	N/A
2006045019	12/2005	WO	N/A
2006078691	12/2005	WO	N/A
2006095188	12/2005	WO	N/A

2006137919	12/2005	WO	N/A
2007035359	12/2006	WO	N/A
2007051092	12/2006	WO	N/A
2007068933	12/2006	WO	N/A
2008024825	12/2007	WO	N/A
2008133765	12/2007	WO	N/A
2009027667	12/2008	WO	N/A
2009106803	12/2008	WO	N/A
2009114928	12/2008	WO	N/A
2009141613	12/2008	WO	N/A
2009141615	12/2008	WO	N/A
2009143169	12/2008	WO	N/A
2009150416	12/2008	WO	N/A
2011008718	12/2010	WO	N/A
2011053972	12/2010	WO	N/A
2011069024	12/2010	WO	N/A
2011087861	12/2010	WO	N/A
2011095810	12/2010	WO	N/A
2011095942	12/2010	WO	N/A
2011106463	12/2010	WO	N/A
2011142768	12/2010	WO	N/A
2012058207	12/2011	WO	N/A
2012080443	12/2011	WO	N/A
2012174265	12/2011	WO	N/A
2013011282	12/2012	WO	N/A
2013116549	12/2012	WO	N/A
2013119423	12/2012	WO	N/A
2014058495	12/2013	WO	N/A
2014107675	12/2013	WO	N/A
2015134802	12/2014	WO	N/A
2016081881	12/2015	WO	N/A
2017084898	12/2016	WO	N/A
2017202793	12/2016	WO	N/A
2018121444	12/2017	WO	N/A
2019217596	12/2018	WO	N/A
2020023603	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report for PCT/US23/60026, Jul. 3, 2023. cited by applicant

Saverskiy et al. “Cargo and Container X-Ray Inspection with Intra-Pulse Multi-Energy Method for Material Discrimination” Physics Procedia, vol. 66, pp. 232-241, ISSN 1875-3892, Jun. 18, 2015 [retrieved on May 9, 2023]. Retrieved from: <URL: <https://www.sciencedirect.com/science/article/pii/S1875389215001832>>. cited by applicant

International Search Report for PCT/US10/35048; Rapiscan Security Products, Inc.; Feb. 8, 2012. cited by applicant

Singh S et al., “Explosives detection systems (EDS) for aviation security”, Signal Processing, Elsevier Science Publishers B.V. Amsterdam, NL, vol. 83, No. 1, Jan. 1, 2003, pp. 31-55, XP027139545, ISSN: 0165-1684. cited by applicant

International Search Report PCT/US2011/025969, mailed on Aug. 1, 2011, Rapiscan Systems Inc. cited by applicant

International Search Report and Written Opinion for PCT/US2010/041757, Oct. 12, 2010. cited by applicant

International Search Report for PCT/US10/58809; Rapiscan Systems Inc.; Apr. 19, 2011. cited by applicant

International Search Report for PCT/GB2009/001250, Mar. 2, 2010, Rapiscan Security Products Inc. cited by applicant

“Mobile X-Ray Inspection Systems” Internet citation Feb. 12, 2007, pp. 1-2, XP007911046 Retrieved from the Internet:
URL:<http://web.archive.org/web/20070212000928/http://www.bombdetection.co-m/cat.sub--details.php?catid=20> [retrieved on Jan. 6, 2010]. cited by applicant

Molchanov P A et al: 'Nanosecond gated optical sensors for ocean optic applications' Sensors Applications Symposium, 2006. Proceedings of The 2006 IEEE Houston, Texas, USA Feb. 7-9, 2006, Piscataway, NJ, USA, IEEE, Feb. 7, 2006 (Feb. 7, 2006) , pp. 147-150, XP010917671 ISBN: 978-0-7803-9580-0. cited by applicant

International Search Report PCT/US2012/024184, mailed on Jul. 27, 2012, Rapiscan Systems Inc. cited by applicant

International preliminary report on patentability PCT/US2012/024184, issued on Aug. 13, 2013, Rapiscan Systems Inc. cited by applicant

International Search Report for PCT/US2010/061908, mailed on Apr. 2, 2012, Rapiscan Systems, Inc. cited by applicant

International Search Report for PCT/GB2006/000859, mailed on May 19, 2006, Corus UK Ltd. cited by applicant

Chou, C, “Fourier coded-aperture imaging in nuclear medicine”, IEEE Proc. Sci. Meas. Technol., vol. 141. No. 3, May 1994, pp. 179-184. cited by applicant

International Search Report and Written Opinion of the International Searching Authority, PCT/US2005/011382, Oct. 21, 2005. cited by applicant

European Patent Office, International Search Report, International Application No. PCT/US99/28266, dated Sep. 6, 2000, 3 pages. cited by applicant

International Preliminary Report on Patentability, PCT/US2005/011382, dated Oct. 19, 2006, 7 pages. cited by applicant

International Preliminary Examining Authority-US, International Preliminary Examination Report, PCT/US1998/018642, dated Aug. 30, 1999, 4 pages. cited by applicant

Mertz, L.N., et al., “Rotational aperture synthesis for x rays”, Journal. Optical Society of America, vol. 3, Dec. 1986, pp. 2167-2170. cited by applicant

International Search Report for PCT/GB2009/001277, Jul. 20, 2010, Rapiscan Systems Inc. cited by applicant

International Search Report PCT/GB2009/000515, Feb. 23, 2010, Rapiscan Security Products, Inc. cited by applicant

International Search Report PCT/GB2009/001444, Apr. 6, 2010, Rapiscan Security Products. cited by applicant

International Search Report for PCT/GB2009/000556, Feb. 19, 2010, Rapiscan Security Products, Inc. cited by applicant

International Search Report, PCT/US2007/066936; dated: Sep. 30, 2008, 5 pages. cited by applicant

International Search Report, PCT/US1998/18642, dated Jul. 7, 1999, 4 pages. cited by applicant

International Search Report, PCT/US1999/028035, dated Sep. 15, 2000, 6 pages. cited by applicant

Written Opinion of the International Searching Authority, PCT/US2007/066936, dated Sep. 30, 2008, 7 pages. cited by applicant

International Search Report for PCT/GB2009/000497, Jan. 22, 2010. cited by applicant

Misso et al., “New developments in radiation detectors and electron multipliers”, 1964, IEEE

Transactions on Nuclear Science pp. 72-75. cited by applicant

International Search Report for PCT/GB2009/001275, Jul. 24, 2009, Rapiscan Security Products Inc. cited by applicant

Keevil, S.V., Lawinski, C.P. and Morton, E.J., 1987, "Measurement of the performance characteristics of anti-scatter grids.", *Phys. Med. Biol.*, 32(3), 397- 403. cited by applicant

Morton, E.J., Webb, S., Bateman, J.E., Clarke, L.J. and Shelton, C.G., 1990, "Three-dimensional x-ray micro-tomography for medical and biological applications.", *Phys. Med. Biol.*, 35(7), 805-820. cited by applicant

Morton, E.J., Swindell, W., Lewis, D.G. and Evans, P.M., 1991, "A linear array scintillation-crystal photodiode detector for megavoltage imaging.", *Med. Phys.*, 18(4), 681-691. cited by applicant

Morton, E.J., Lewis, D.G. and Swindell, W., 1988, "A method for the assessment of radiotherapy treatment precision", *Brit. J. Radiol.*, Supplement 22, 25. cited by applicant

Swindell, W., Morton, E.J., Evans, P.M. and Lewis, D.G., 1991, "The design of megavoltage projection imaging systems: some theoretical aspects.", *Med. Phys.*, 18(5), 855-866. cited by applicant

Morton, E.J., Evans, P.M., Ferraro, M., Young, E.F. and Swindell, W., 1991, "A video frame store facility for an external beam radiotherapy treatment simulator.", *Brit. J. Radiol.*, 64, 747-750. cited by applicant

Antonuk, L.E., Yorkston, J., Kim, C.W., Huang, W., Morton, E.J., Longo, M.J. and Street, R.A., 1991, "Light response characteristics of amorphous silicon arrays for megavoltage and diagnostic imaging.", *Mat. Res. Soc. Sym. Proc.*, 219, 531-536. cited by applicant

Yorkston, J., Antonuk, L.E., Morton, E.J., Boudry, J., Huang, W., Kim, C.W., Longo, M.J. and Street, R.A., 1991, "The dynamic response of hydrogenated amorphous silicon imaging pixels.", *Mat. Res. Soc. Sym. Proc.*, 219, 173-178. cited by applicant

Evans, P.M., Gildersleve, J.Q., Morton, E.J., Swindell, W., Coles, R., Ferraro, M., Rawlings, C., Xiao, Z.R. and Dyer, J., 1992, "Image comparison techniques for use with megavoltage imaging systems.", *Brit. J. Radiol.*, 65, 701-709. cited by applicant

Morton, E.J., Webb, S., Bateman, J.E., Clarke, L.J. and Shelton, C.G., 1989, "The development of 3D x-ray micro-tomography at sub 100Å resolution with medical, industrial and biological applications.", Presentation at IEE colloquium "Medical scanning and imaging techniques of value in non-destructive testing", London, Nov. 3, 1989. cited by applicant

Antonuk, L.E., Boudry, J., Huang, W., McShan, D.L., Morton, E.J., Yorkston, J., Longo, M.J. and Street, R.A., 1992, "Demonstration of megavoltage and diagnostic x-ray imaging with hydrogenated amorphous silicon arrays.", *Med. Phys.*, 19(6), 1455-1466. cited by applicant

Gildersleve, J.Q., Swindell, W., Evans, P.M., Morton, E.J., Rawlings, C. and Dearnaley, D.P., 1991, "Verification of patient positioning during radiotherapy using an integrated megavoltage imaging system.", in "Tumour Response Monitoring and Treatment Planning", Proceedings of the International Symposium of the W. Vaillant Foundation on Advanced Radiation Therapy, Munich, Germany, Ed A. Breit (Berlin: Springer), 693-695. cited by applicant

Lewis, D.G., Evans, P.M., Morton, E.J., Swindell, W. and Xiao, X.R., 1992, "A megavoltage CT scanner for radiotherapy verification.", *Phys. Med. Biol.*, 37, 1985-1999. cited by applicant

Antonuk, L.E., Boudry, J., Kim, C.W., Longo, M.J., Morton, E.J., Yorkston, J. and Street, R.A., 1991, "Signal, noise and readout considerations in the development of amorphous silicon photodiode arrays for radiotherapy and diagnostic x-ray imaging.", *SPIE vol. 1443 Medical Imaging V: Image Physics*, 108-119. cited by applicant

Antonuk, L.E., Yorkston, J., Huang, W., Boudry, J., Morton, E.J., Longo, M.J. and Street, R.A., 1992, "Radiation response characteristics of amorphous silicon arrays for megavoltage radiotherapy imaging.", *IEEE Trans. Nucl. Sci.*, 39, 1069-1073. cited by applicant

Antonuk, L.E., Yorkston, J., Huang, W., Boudry, J., Morton, E.J., Longo, M.J. and Street, R.A., 1992, "Factors affecting image quality for megavoltage and diagnostic x-ray a-Si:H imaging

arrays.”, Mat. Res. Soc. Sym. Proc., 258, 1069-1074. cited by applicant

Antonuk, L.E., Boudry, J., Yorkston, J., Morton, E.J., Huang, W. and Street, R.A., 1992, “Development of thin-film, flat-panel arrays for diagnostic and radiotherapy imaging.”, SPIE vol. 1651, Medical Imaging VI: Instrumentation, 94-105. cited by applicant

Yorkston, J., Antonuk, L.E., Seraji, N., Boudry, J., Huang, W., Morton, E.J., and Street, R.A., 1992, “Comparison of computer simulations with measurements from a-Si:H imaging arrays.”, Mat. Res. Soc. Sym. Proc., 258, 1163-1168. cited by applicant

Morton, E.J., Antonuk, L.E., Berry, J.E., Boudry, J., Huang, W., Mody, P., Yorkston, J. and Longo, M.J., 1992, “A Camac based data acquisition system for flat-panel image array readout”, Presentation at IEEE Nuclear Science Symposium, Orlando, Oct. 25-31, 1992. cited by applicant

Antonuk, L.E., Yorkston, J., Huang, W., Boudry, J., Morton, E.J. and Street, R.A., 1993, “Large area, flat-panel a-Si:H arrays for x-ray imaging.”, SPIE vol. 1896, Medical Imaging 1993: Physics of Medical Imaging, 18-29. cited by applicant

Morton, E.J., Antonuk, L.E., Berry, J.E., Huang, W., Mody, P. and Yorkston, J., 1994, “A data acquisition system for flat-panel imaging arrays”, IEEE Trans. Nucl. Sci., 41(4), 1150-1154. cited by applicant

Antonuk, L.E., Boudry, J., Huang, W., Lam, K.L., Morton, E.J., TenHaken, R.K., Yorkston, J. and Clinthorne, N.H., 1994, “Thin-film, flat-panel, composite imagers for projection and tomographic imaging”, IEEE Trans. Med. Im., 13(3), 482-490. cited by applicant

Gildersleve, J., Dearnaley, D., Evans, P., Morton, E.J. and Swindell, W., 1994, “Preliminary clinical performance of a scanning detector for rapid portal imaging”, Clin. Oncol., 6, 245-250. cited by applicant

Hess, R., De Antonis, P., Morton, E.J. and Gilboy, W.B., 1994, “Analysis of the pulse shapes obtained from single crystal CdZnTe radiation detectors”, Nucl. Inst. Meth., A353, 76-79. cited by applicant

Deantonis, P., Morton, E.J., T. Menezes, 1996, “Measuring the bulk resistivity of CdZnTe single crystal detectors using a contactless alternating electric field method”, Nucl. Inst. Meth., A380, 157-159. cited by applicant

Deantonis, P., Morton, E.J., Podd, F., 1996, “Infra-red microscopy of CdZnTe radiation detectors revealing their internal electric field structure under bias”, IEEE Trans. Nucl. Sci., 43(3), 1487-1490. cited by applicant

Tavora, L.M.N., Morgado, R.E., Estep, R.J., Rawool-Sullivan, M., Gilboy, W.B. and Morton, E.J., 1998, “One-sided imaging of large, dense, objects using the 511 keV photons from induced pair production”, IEEE Trans. Nucl. Sci., 45(3), 970-975. cited by applicant

Morton, E.J., 1995, “Archaeological potential of computerised tomography”, Presentation at IEE Colloquium on “NDT in archaeology and art”, London, May 25, 1995. cited by applicant

Tavora, L.M.N. and Morton, E.J., 1998, “Photon production using a low energy electron expansion of the EGS4 code system”, Nucl. Inst. Meth., B143, 253-271. cited by applicant

Patel, D.C. and Morton, E.J., 1998, “Analysis of improved adiabatic pseudo- domino logic family”, Electron. Lett., 34(19), 1829-1830. cited by applicant

Kundu, A and Morton, E.J., 1999, “Numerical simulation of argon-methane gas filled proportional counters”, Nucl. Inst. Meth., A422, 286-290. cited by applicant

Luggar, R.D., Key, M.J., Morton, E.J. and Gilboy, W.B., 1999, “Energy dispersive X-ray scatter for measurement of oil/water ratios”, Nucl. Inst. Meth., A422, 938-941. cited by applicant

Morton, E.J., Crockett, G.M., Sellin, P. J. and DeAntonis, P., 1999, “The charged particle response of CdZnTe radiation detectors”, Nucl. Inst. Meth., A422, 169-172. cited by applicant

Morton, E.J., Clark, R.J. and Crowley, C., 1999, “Factors affecting the spectral resolution of scintillation detectors”, Nucl. Inst. Meth., A422, 155-158. cited by applicant

Morton, E.J., Caunt, J.C., Schoop, K., Swinhoe, M., 1996, “A new handheld nuclear material analyser for safeguards purposes”, Presentation at INMM annual meeting, Naples, Florida, Jul.

1996. cited by applicant

Hepworth, S., McJury, M., Oldham, M., Morton, E.J. and Doran, S.J., 1999, "Dose mapping of inhomogeneities positioned in radiosensitive polymer gels", Nucl. Inst. Meth., A422, 756-760. cited by applicant

Morton, E.J., Luggar, R.D., Key, M.J., Kundu, A., Tavora, L.M.N. and Gilboy, W.B., 1999, "Development of a high speed X-ray tomography system for multiphase flow imaging", IEEE Trans. Nucl. Sci., 46 III(1), 380-384. cited by applicant

Tavora, L.M.N., Morton, E.J., Santos, F.P. and Dias, T.H.V.T., 2000, "Simulation of X-ray tubes for imaging applications", IEEE Trans. Nucl. Sci., 47, 1493-1497. cited by applicant

Tavora, L.M.N., Morton, E.J. and Gilboy, W.B., 2000, "Design considerations for transmission X-ray tubes operated at diagnostic energies", J. Phys. D: Applied Physics, 33(19), 2497-2507. cited by applicant

Morton, E.J., Hossain, M.A., DeAntonis, P. and Ede, A.M.D., 2001, "Investigation of Au-CdZn Te contacts using photovoltaic measurements", Nucl. Inst. Meth., A458, 558-562. cited by applicant

Ede, A.M.D., Morton, E.J. and DeAntonis, P., 2001, "Thin-film CdTe for imaging detector applications", Nucl. Inst. Meth., A458, 7-11. cited by applicant

Tavora, L.M.N., Morton, E.J. and Gilboy, W.B., 2001, "Enhancing the ratio of fluorescence to bremsstrahlung radiation in X-ray tube spectra", App. Rad. and Isotopes, 54(1), 59-72. cited by applicant

Menezes, T. and Morton, E.J., 2001, "A preamplifier with digital output for semiconductor detectors", Nucl. Inst. Meth. A., A459, 303-318. cited by applicant

Johnson, D.R., Kyriou, J., Morton, E.J., Clifton, A.C. Fitzgerald, M. and MacSweeney, J.E., 2001, "Radiation protection in interventional radiology", Clin. Rad., 56(2), 99-106. cited by applicant

Tavora, L.M.N., Gilboy, W.B. and Morton, E.J., 2001, "Monte Carlo studies of a novel X-ray tube anode design", Rad. Phys. and Chem., 61, 527-529. cited by applicant

"Morton, E.J., 1998, "Is film dead: the flat plate revolution", Keynote Talk, IPEM Annual Conference, Brighton, 14-17 Sept, 1998"}]. cited by applicant

Luggar, R.D., Morton, E.J., Jenneson, P.M. and Key, M.J., 2001, "X-ray tomographic imaging in industrial process control", Rad. Phys. Chem., 61, 785-787. cited by applicant

Luggar, R.D., Morton, E.J., Key, M.J., Jenneson, P.M. and Gilboy, W.B., 1999, "An electronically gated multi-emitter X-ray source for high speed tomography", Presentation at SPIE Annual Meeting, Denver, Jul. 19-23, 1999. cited by applicant

Gregory, P.J., Hutchinson, D.J., Read, D.B., Jenneson, P.M., Gilboy, W.B. and Morton, E.J., 2001, "Non-invasive imaging of roots with high resolution X-ray microtomography", Plant and Soil, 255(1), 351-359. cited by applicant

Kundu, A., Morton, E.J., Key, M.J. and Luggar, R.D., 1999, "Monte Carlo simulations of microgap gas-filled proportional counters", Presentation at SPIE Annual Meeting, Denver, Jul. 19-23, 1999. cited by applicant

Hossain, M.A., Morton, E.J., and Ozsan, M.E., 2002, "Photo-electronic investigation of CdZn Te spectral detectors", IEEE Trans. Nucl. Sci, 49(4), 1960-1964. cited by applicant

Panman, A., Morton, E.J., Kundu, A and Sellin, P.J., 1999, "Optical Monte Carlo transport in scintillators", Presentation at SPIE Annual Meeting, Denver, Jul. 19-23, 1999. cited by applicant

Jenneson, P.M., Gilboy, W.B., Morton, E.J., and Gregory, P.J., 2003, "An X-ray micro-tomography system optimised for low dose study of living organisms", App. Rad. Isotopes, 58, 177-181. cited by applicant

Key, M.J., Morton, E.J., Luggar, R.D. and Kundu, A., 2003, "Gas microstrip detectors for X-ray tomographic flow imaging", Nucl. Inst. Meth., A496, 504-508. cited by applicant

Jenneson, P.M., Luggar, R.D., Morton, E.J., Gundogdu, O, and Tuzun, U, 2004, "Examining nanoparticle assemblies using high spatial resolution X-ray microtomography", J. App. Phys, 96(5), 2889-2894. cited by applicant

Tavora, L.M., Gilboy, W.B. and Morton, E.J., 2000, "Influence of backscattered electrons on X-ray tube output", Presentation at SPIE Annual Meeting, San Diego, 30 July - Aug. 3, 2000. cited by applicant

Wadson, N., Morton, E.J., and Lionheart, W.B., 2010, "Scatter in an uncollimated x-ray CT machine based on a Geant4 Monte Carlo simulation", SPIE Medical Imaging 2010: Physics of Medical Imaging, Feb. 15-18, 2010, San Diego, USA. cited by applicant

Morton, E.J., 2010, "Position sensitive detectors in security: Users perspective", Invited talk, STFC meeting on position sensitive detectors, RAL, May 2010. cited by applicant

Third Party Submission Under 37 CFR 1.290 for U.S. Appl. No. 15/954,853, filed Apr. 19, 2019. cited by applicant

Domingo Mery, "Computer Vision for X-Ray Testing: Imaging, Systems, Image Databases, and Algorithms", Springer International Publishing Switzerland 2015 (Year: 2015). cited by applicant

Chen C et al., "Front-end electronics for the CDF-II time-of-flight system", IEEE Transactions On Nuclear Science, IEEE Service Center, New York, Ny, US, (20031201), vol. 50, No. 6, doi:10.1109/TNS.2003.820632, ISSN 0018-9499, pp. 2486-2490, XP011106678. cited by applicant

[XAI] - Wen Xianfei et al., "Measuring the scintillation decay time for different energy deposited by [gamma]-rays and neutrons in a Cs2LiYCl6:Ce3+detector", Nuclear Instruments & Methods in Physics Research. Section A: Accelerators, Spectrometers, Detectors, and Associated Equipment, Elsevier Bv * North-Holland, NL, (20170209), vol. 853, doi:10.1016/J.NIMA.2017.02.019, ISSN 0168-9002, pp. 9-15, XP029936730. cited by applicant

Kyle Polack et al. "Dual-Particle Imager for Standoff Detection of Special Nuclear Material", IEEE Nuclear Science Symposium Conference Record, Oct. 23, 2011, pp. 1494-1500, IEEE. cited by applicant

Soundara-Pandian L et al., "Lithium Alkaline Halides-Next Generation of Dual Mode Scintillators", IEEE Transactions On Nuclear Science, IEEE Service Center, New York, Ny, US, vol. 63, No. 2, doi:10.1109/TNS.2016.2535355, ISSN 0018-9499, (20160401), pp. 490-496, (20160418), XP011606934. cited by applicant

Primary Examiner: Gunberg; Edwin C

Attorney, Agent or Firm: Novel IP

Background/Summary

CROSS-REFERENCE (1) The present application relies on U.S. Patent Provisional Application No. 63/369,386, titled "Methods and Systems for Performing On-The-Fly Automatic Calibration Adjustments of X-Ray Inspection Systems", and filed on Jul. 26, 2022, for priority, which is herein incorporated by reference in its entirety.

FIELD

(1) The present specification is related generally to the field of X-ray inspection. More specifically, the present specification is related to a method of enabling automatic adjustment of calibration data of X-ray inspection systems while scanning an object under examination or inspection.

BACKGROUND

(2) Most modern high energy (2.5-9 MeV) transmission X-ray non-intrusive inspection systems offer some degree of cargo composition estimation by using interlaced dual energy pulses. By comparing the radiographic images of the low energy (3-5 MeV) and high energy (6-9 MeV) of the same container, different regions inside it can be characterized into four broad categories of materials: organic, inorganic (such as soil, fertilizer, salts, aluminum), metallic (such as steel, scrap

metal), and High-Z materials (such as materials having an atomic number at or above tungsten).

(3) Unlike the low energy inspection systems used for parcel and luggage screening where the prevailing process is the photoelectric effect, at high energies, the Compton effect dominates the interaction of X-rays with matter. The Compton process is weakly dependent on the effective atomic number of the material, making an accurate assessment difficult.

(4) In addition to the challenges brought by physics, there are other challenges specific to the cargo screening such as, but not limited to: a) cargo clutter, where multiple optically dense objects overlap in the image; b) X-ray source stability (in terms of maintaining a consistent and/or predictable value for dose and energy) and detector stability, both of which may lead to calibration drift; c) energy-dependent X-ray scatter, where low and high energy beams produce different blur in the image and d) obscured objects of interest (such as high-Z shielding, weapons, or smuggled drug packages), which are often small compared to the rest of the imaged cargo.

(5) Dual energy material separation methods involve a calibration process in which a set of materials with known atomic numbers and densities such as plastic, aluminum, steel, lead, tungsten, with different thicknesses are placed between the source and detectors, a scan is performed and responses of detectors for the high and low energy beam components are recorded. In normal operation mode, the inverse process takes place. Detectors record the high and low energy response to objects in the beam, and then algorithms determine the closest candidate, based on an effective atomic number (Z_{eff}), in terms of material type and thickness. An uncertainty is associated with the determination of the material type, depending on the system geometry, X-ray flux and energies, and the number of samples (pulse pairs) used to make the determination.

(6) Unfortunately, materials separation calibration does not remain stable over the lifetime of an X-ray inspection system. Changes often appear that make the Z_{eff} calculation unreliable. In some conditions, the drift may even happen between two scans taken minutes apart.

(7) Therefore, there is a need for a method that enables automatic modulation of calibration data for an X-ray system during scanning of an object under examination or inspection.

SUMMARY

(8) The following embodiments and aspects thereof are described and illustrated in conjunction with systems, tools and methods, which are meant to be exemplary and illustrative, and not limiting in scope. The present application discloses numerous embodiments.

(9) In some embodiments, the present specification discloses a method of performing an on-the-fly adjustment of calibration data corresponding to an X-ray inspection system that comprises an X-ray source in data communication with a computing device, the method being implemented in the computing device having one or more physical processors programmed with a plurality of program instructions that, when executed by the one or more physical processors, cause the computing device to perform the method, the method comprising: generating, by the X-ray inspection system, a first plurality of calibration data, wherein the first plurality of calibration data is stored in a memory associated with the computing device; generating, using the X-ray inspection system, data representative of X-ray scan images of a cargo container; segmenting data representative of X-ray scan images into a plurality of regions of interest, wherein at least some of the plurality of regions of interest contain only one type of material having at least one of a known effective atomic number (Z_{eff}) or a known HS code; determining, using the first plurality of calibration data, first data indicative of Z_{eff} of the at least some of the plurality of regions of interest; comparing the first data with second data indicative of Z_{eff} corresponding to the known Z_{eff} or known HS code; adjusting the first plurality of calibration data to generate a second plurality of calibration data if the first and second data differ more than a predefined threshold; and using the second plurality of calibration data to process the data representative of X-ray scan images to generate and display X-ray scan images.

(10) Optionally, the method further comprises calculating a Z_{eff} value for each pixel in the X-ray scan images; applying a smoothing algorithm to the Z_{eff} values of multiple pixels, associated with

each of the plurality of regions of interest, in order to generate smoothed Zeff-based regions of interest; applying a predefined color map to the smoothed Zeff-based regions of interest; and displaying the color-coded X-ray scan images.

(11) Optionally, the Zeff value for each pixel is calculated using the second plurality of calibration data if the first and second data differ more than the predefined threshold, and wherein the Zeff value for each pixel is calculated using the first plurality of calibration data if the first and second data do not differ more than the predefined threshold.

(12) Optionally, the smoothing algorithm includes any one of simple running averages, shape preserving smoothing filters, image-guided segmentation or K-means clustering.

(13) Optionally, the at least some of the plurality of regions of interest correspond to empty or unobstructed regions of the cargo container.

(14) Optionally, the at least some of the plurality of regions of interest correspond to at least one of a volume encompassing only a top surface of the cargo container, a volume encompassing only a front end of the cargo container, or a volume encompassing only a rear end of the cargo container.

(15) Optionally, the first plurality of calibration data is generated by: placing, between a radiation source and detectors of the X-ray inspection system, a set of materials with known atomic numbers and densities, wherein the set of materials have different thicknesses; and recording responses of the detectors for each of the set of materials.

(16) Optionally, the X-ray scan images correspond to low and high energies of the radiation source.

(17) In some embodiments, the present specification is directed towards an X-ray inspection system comprising: a radiation source; an array of detectors; a computing device in data communication with the radiation source and the array of detectors, wherein the computing device has one or more physical processors programmed with a plurality of program instructions that, when executed by the one or more physical processors, cause the computing device to generate, using the X-ray inspection system, a first plurality of calibration data, wherein the first plurality of calibration data are stored in a memory associated with the computing device; generate, using the X-ray inspection system, X-ray scan images of a cargo container; segment the X-ray scan images into a plurality of regions of interest, wherein each of the plurality of regions of interest contain a single type of a material associated with a known atomic number or a single type of a material associated with a known HS code; determine, using the first plurality of calibration data, first data indicative of a first effective atomic number of the single type of the material associated with the known atomic number or a second effective atomic number of the single type of the material associated with the known HS code for each of the plurality of regions of interest; compare the first data with second data indicative of a known effective atomic number corresponding to the material associated with the known atomic number or a known effective atomic number corresponding to the material associated with the known HS code; adjust the first plurality of calibration data to generate a second plurality of calibration data if the first and second data differ significantly; and replace the first plurality of calibration data by the second plurality of calibration data in the memory.

(18) Optionally, the plurality of program instructions, when executed by the one or more physical processors, further cause the computing device to: calculate an effective atomic number value for each pixel in the X-ray scan images; apply a smoothing algorithm to the effective atomic number values of multiple pixels, associated with each of the plurality of regions of interest, in order to generate smoothed effective atomic number-based regions of interest; apply a predefined color map to the smoothed effective atomic number-based regions of interest; and display the color coded X-ray scan images.

(19) Optionally, the effective atomic number value for each pixel is calculated using the second plurality of calibration data if the first and second data differ significantly, and wherein the effective atomic number value for each pixel is calculated using the first plurality of calibration data if the first and second data do not differ significantly.

(20) Optionally, the smoothing algorithm includes any one of simple running averages, shape

preserving smoothing filters, image-guided segmentation or K-means clustering.

(21) Optionally, at least some of the plurality of regions of interest are empty or unobstructed.

(22) Optionally, at least some of the plurality of regions of interest contain correspond to at least one of a volume encompassing only a top surface of the cargo container, a volume encompassing only a front end of the cargo container, or a volume encompassing only a rear end of the cargo container.

(23) Optionally, the first plurality of calibration data is generated by: placing, between a radiation source and detectors of the X-ray inspection system, a set of materials with known atomic numbers and densities, wherein the set of materials have different thicknesses; and recording response of the detectors for each of the set of materials.

(24) Optionally, the X-ray scan images correspond to low and high energies of the radiation source.

(25) The aforementioned and other embodiments of the present specification shall be described in greater depth in the drawings and detailed description provided below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate various embodiments of systems, methods, and embodiments of various other aspects of the disclosure. Any person with ordinary skills in the art will appreciate that the illustrated element boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one example of the boundaries. It may be that in some examples one element may be designed as multiple elements or that multiple elements may be designed as one element. In some examples, an element shown as an internal component of one element may be implemented as an external component in another and vice versa. Furthermore, elements may not be drawn to scale. Non-limiting and non-exhaustive descriptions are described with reference to the following drawings. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating principles.

(2) FIG. 1 is a schematic drawing of an exemplary LINAC-based high-energy X-ray cargo inspection system that may implement a method described in the present specification in an embodiment;

(3) FIG. 2A shows segmented regions of a radiographic scan image of a cargo container, in accordance with some embodiments of the present specification;

(4) FIG. 2B shows a radiographic scan image of a container housing bananas, in accordance with some embodiments of the present specification; and

(5) FIG. 3 is a flowchart of a plurality of exemplary steps of a method of processing X-ray scan images of an object under inspection for materials classification and on-the-fly automatic adaptive calibration, in accordance with some embodiments of the present specification.

DETAILED DESCRIPTION

(6) The present specification is directed towards multiple embodiments. The following disclosure is provided in order to enable a person having ordinary skill in the art to practice the invention.

Language used in this specification should not be interpreted as a general disavowal of any one specific embodiment or used to limit the claims beyond the meaning of the terms used therein. The general principles defined herein may be applied to other embodiments and applications without departing from the spirit and scope of the invention. Also, the terminology and phraseology used is for the purpose of describing exemplary embodiments and should not be considered limiting. Thus, the present invention is to be accorded the widest scope encompassing numerous alternatives, modifications and equivalents consistent with the principles and features disclosed. For purpose of clarity, details relating to technical material that is known in the technical fields related to the invention have not been described in detail so as not to unnecessarily obscure the present invention.

(7) In various embodiments, a computing device includes an input/output controller, at least one communications interface and system memory. The system memory includes at least one random access memory (RAM) and at least one read-only memory (ROM). These elements are in communication with a central processing unit (CPU) to enable operation of the computing device. In various embodiments, the computing device may be a conventional standalone computer or alternatively, the functions of the computing device may be distributed across multiple computer systems and architectures.

(8) In some embodiments, execution of a plurality of sequences of programmatic instructions or code enable or cause the CPU of the computing device to perform various functions and processes. In alternate embodiments, hard-wired circuitry may be used in place of, or in combination with, software instructions for implementation of the processes of systems and methods described in this application. Thus, the systems and methods described are not limited to any specific combination of hardware and software.

(9) In the description and claims of the application, each of the words “comprise”, “include”, “have”, “contain”, and forms thereof, are not necessarily limited to members in a list with which the words may be associated. Thus, they are intended to be equivalent in meaning and be open-ended in that an item or items following any one of these words is not meant to be an exhaustive listing of such item or items, or meant to be limited to only the listed item or items. It should be noted herein that any feature or component described in association with a specific embodiment may be used and implemented with any other embodiment unless clearly indicated otherwise.

(10) It must also be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural references unless the context dictates otherwise. Although any systems and methods similar or equivalent to those described herein can be used in the practice or testing of embodiments of the present disclosure, the preferred, systems and methods are now described.

(11) “On-the-fly adjustment” refers to performing an adjustment to calibration data 1) immediately after a scan has occurred (first scan) and before an immediately subsequent scan is performed, 2) based on the scan data that was just acquired from the first scan, and 3) without manual or human intervention.

Overview

(12) FIG. 1 illustrates an exemplary LINAC-based high-energy X-ray cargo inspection system that may be configured and used, in an embodiment, to implement the method(s) described in the present specification. As shown, the cargo inspection system **130** comprises a high-energy radiation source **105** for irradiating an object under inspection **110** with a vertically divergent fan beam of radiation **115**. The high-energy radiation source **105** may be, but is not limited to, a linear accelerator (LINAC) or Betatron. In embodiments, the LINAC or any other source provides a radiation dose sufficient for imaging containers and cargo. In an embodiment, the energy and dose output of the LINAC or any other source ranges from 750 keV to 10 MeV and 0.07 Gy/min to 15 Gy/min, respectively.

(13) The choice of source type, source intensity, and energy output depends upon the sensitivity of the detectors, the radiographic density of the cargo positioned in the space between the source and detectors, radiation safety considerations, and operational requirements, such as the inspection speed. One of ordinary skill in the art would appreciate the factors that need to be considered when selecting a radiation source type depend upon inspection requirements. In one embodiment, where the object under inspection **110** is a large-sized container or car that highly attenuates the X-ray beam, the radiation may be from an X-ray source operating at an energy ranging from approximately 750 keV and even up to 10 MeV or more. In various embodiments, the object under inspection **110** may be a vehicle, truck, rail car or other containers for carrying cargo, passenger luggage or general belongings.

(14) The cargo inspection system **130** further comprises a detector array **120**, which is preferably

positioned behind the object under inspection **110** and is configured to and used to detect radiation transmitted through the object under inspection **110**. The detectors **120**, in an embodiment, are formed by a stack of crystals that generate analog signals when X-rays impinge upon them, with the signal strength proportional to the amount of beam attenuation in the object under inspection **110**. In one embodiment, the X-ray beam detector arrangement consists of a linear array of solid-state detectors of the crystal-diode type. A typical arrangement uses cadmium tungstate scintillating crystals to absorb the X-rays transmitted through the object under inspection **110** and to convert the absorbed X-rays into photons of visible light. Crystals such as bismuth germanate, sodium iodide or other suitable crystals may be alternatively used as known to a person of ordinary skill in the art. The crystals can be directly coupled to a suitable detector, such as a photodiode or photo-multiplier. The detector photodiodes may be linearly arranged, which through unity-gain devices, provide advantages over photo-multipliers in terms of operating range, linearity and detector-to-detector matching. In another embodiment, an area detector is used as an alternative to linear array detectors. Such an area detector may be a scintillating strip, such as cesium iodide or other materials known in the art, viewed by a suitable camera or optically coupled to a charge-coupled device (CCD).

(15) It would be apparent to persons of skill in the art that the cargo inspection system **130** shown in FIG. 1 is just one example of an inspection system employing high-energy X-ray sources such as, but not limited to LINAC or Betatron. In some embodiments, the radiation source **105** uses interlaced dual energy pulses in order to generate X-ray scan images of low energy (3-5 MeV) and high energy (6-9 MeV) of the object under inspection **110**.

Calibration Drift

(16) The present specification recognizes that calibration data for materials separation does not remain stable over the lifetime of an X-ray inspection system. Changes in calibration, also referred to as “calibration drift” often appear and render the Zeff (effective atomic number) calculation unreliable for effective materials separation data. In some conditions, the drift may happen even between two scans taken minutes apart. The following sources, factors, causes or variables may all contribute to calibration drift. The X-ray source itself may be a contributing cause of calibration drift. Dose drift may be induced by beam current or beam energy changes. These include magnet changes (Betatron), automatic frequency control (AFC) shift, and wave guide temperature variation. Beam formation mechanics, which may include collimator alignment casting shadows, detector stack boom oscillations, detector collimator fins shift may also contribute to calibration drift. In addition, X-ray detectors may have problems with photodiode dark current and scintillator light yield, which contribute to calibration drift.

(17) More than one cause of calibration shift may occur at the same time. However, only a few of the sources, factors, causes or variables of calibration drift have built-in mechanisms for compensation such as, for example, reference detectors and dark current measurement at the beginning of a scan or in-between pulses. These sources, factors, causes or variables of calibration drift are generally non-linear and are often linked with each other in complicated ways. Also, full recalibration is time consuming and is only a temporary solution for causes such as temperature excursions or slight magnetic drift (in a Betatron) or AFC drift (in a LINAC).

Automatic Adaptive Calibration

(18) In accordance with aspects of the present specification, instead of parametrizing the contribution of each source, factor, cause or variable to the calibration drift, a phenomenological model is developed in which the system is configured to check an accuracy of the Zeff calculation in almost every scan and in which the system is configured to automatically adjust the calibration data in order to improve scan data, and in particular, materials separation accuracy.

(19) FIG. 2A shows segmented regions of a radiographic scan image **200** of a cargo container **202**, in accordance with some embodiments of the present specification. In a non-limiting exemplary scenario, the image **200** has at least five regions or sections (also referred to as regions or sections

of interest), that is, first region **205**, second region **207**, third region **209**, fourth region **211**, and fifth region **213** for which the material is conventionally known to be steel (for ISO containers, for example). Also, for the at least five regions **205**, **207**, **209**, **211**, **213** the attenuation and angular range (relative to a central axis of an X-ray beam used to scan the cargo container **202**) vary enough. In embodiments, the regions are selected for different purposes. For example, referring to FIG. 2A, and by way of example only, regions **207**, **209**, and **211** are selected for stability since the variation is relatively small compared to regions **205** and **213**, which are selected to have large angular coverage and varying attenuations.

(20) In embodiments, a radiographic scan image, such as the image **200** of the cargo container **202**, is segmented into one or more regions or sections of interest based on a variability of attenuation and an angular range relative to the central axis of the X-ray beam. In some embodiments, the one or more regions or sections of interest are selected such that the expected attenuation (of the regions or sections of interest) varies more than 10% and the angular range or span is greater than 2 degrees relative to the central axis of the X-ray beam.

(21) Containers, such as the cargo container **202**, typically have one or more regions such as, for example, a front end (comprising first region **205**) and a rear end (comprising fifth region **213**) and a top portion (comprising second region **207**, third region **209**, and fourth region **211**) that remain unobstructed. It should be noted that the fifth region **213** and at least the second top region **207** are almost always unobstructed for most cargo containers. The first region or front end **205** is also unobstructed in most cases. The second region **207**, third region **209**, and fourth region **211** are illustrative non-limiting examples of areas or portions in the image **200** that are found to be unobstructed and that can be used for calibration adjustment. These regions will, however, differ on a case-by-case basis depending on how a container is loaded. In some embodiments, a bottom surface or portion **227** of the cargo container **202** may also be at least partially unobstructed (with known container material). Therefore, in various embodiments, an X-ray scan image of a container is analyzed and segmented to identify regions or areas for which material composition is known, wherein such regions or areas of known material composition include portions inside or within the container such as, but not limited to, unobstructed regions (that is, regions not loaded with cargo) of the container, and portions outside the container such as, but not limited to, tires, windshield, vehicle frame around axles and fuel or gas tank.

(22) For example, in one embodiment, the region of interest comprising the top portion (comprising at least one of the second region **207**, third region **209**, and fourth region **211**) of the cargo container **202** is made of only the material that separates the internal volume of the cargo container **202** from the outside environment. In another embodiment, the material that separates the internal volume of the cargo container **202** from the outside environment makes up at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or any numerical increment between 20% and 90% of the region of interest comprising the top portion of the cargo container **202**.

(23) In another example, in one embodiment, the region of interest comprising the front end (comprising first region **205**) of the cargo container **202** is made of only the material that separates the internal volume of the cargo container **202** from the outside environment. In another embodiment, the material that separates the internal volume of the cargo container **202** from the outside environment makes up at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or any numerical increment between 20% and 90% of the region of interest comprising the front end of the cargo container **202**.

(24) In yet another example, in one embodiment, the region of interest comprising the rear end (comprising fifth region **213**) of the cargo container **202** is made of only the material that separates the internal volume of the cargo container **202** from the outside environment. In another embodiment, the material that separates the internal volume of the cargo container **202** from the outside environment makes up at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or any numerical increment between 20% and 90% of the region of interest comprising the rear end of the

cargo container **202**.

(25) In still another example, in one embodiment, the region of interest comprising the bottom surface or portion **227** of the cargo container **202** is made of only the material that separates the internal volume of the cargo container **202** from the outside environment. In another embodiment, the material that separates the internal volume of the cargo container **202** from the outside environment makes up at least 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, or any numerical increment between 20% and 90% of the region of interest comprising the bottom surface or portion of the cargo container **202**.

(26) Also, the first region **205** and fifth region **213** are of interest because of the angular coverage or span of more than 15 degrees off the central axis of the X-ray beam. As known to persons of ordinary skill in the art, due to the physics of X-ray production, the shape of an attenuation curve is expected to change if the energy of the electrons producing the X-rays changes. High energy beams have flux and energy composition roll-off as a function of angle relative to the central axis of the X-ray beam. The roll-off function is different at 4 MeV versus 6 MeV. In some embodiments, the change, drift or modification of the calibration curves is determined by calculating the variation of the roll-off curves.

(27) In accordance with some embodiments, the unobstructed regions, such as second region **207**, third region **209**, and fourth region **211**, are identified and used as data points. In some embodiments, machine learning techniques such as, but not limited to, mask R-CNNs (Region-based Convolutional Neural Networks) or U-Nets are configured to segment the image **200** for more features inside and outside the cargo container **202** suitable for use in automatic calibration adjustments.

(28) In some embodiments, for scans that have the HS code associated with the X-ray image, the cargo inside the container is used to check and adjust, if required, the calibration data. For example, FIG. 2B shows a radiographic scan image **220** of a container **230** housing bananas, in accordance with some embodiments of the present specification. The dotted box **225** shows that the cargo segmentation algorithm successfully identified the cargo region **222** inside the refrigerated ISO container **230**.

(29) In accordance with various embodiments, there are multiple other sections or regions inside the container **230** that can be used to check and, if required, adjust the calibration data. The top edge region **235** of the container **230** is metallic and uncluttered, the back edge **240** of the container **230** is also metallic and spans a large angular range relative to a central axis of an X-ray beam used to scan the container **230**. Inside, the bananas **222** are organic with a large angular and attenuation coverage.

(30) Thus, in embodiments, the present specification is directed towards a radiographic or X-ray image processing method of identifying or segmenting one or more regions or sections of interest of a cargo container for which the materials are known and that have associated large angular and attenuation coverage. In various embodiments, the one or more regions or sections of interest have an associated angular span of more than 2 degrees and an attenuation variation of more than 10% relative to the mean inside each of the one or more regions or sections of interest. Thus, in embodiments, this refers to particular values relative to the values within the region itself. As a non-limiting example, this may represent regions **205** and **213**. Thereafter, the method computationally links the variation of Z_{eff} values of the identified or segmented regions of interest relative to the current calibration and adjusts the parameters of the calibration curves to match with known materials.

(31) In embodiments, the regions of interest may become obstructed by known features and materials, such as, for example, the skin of the cargo container. In order to be a region of interest, a condition is being able to determine (or having a way of knowing) what material is contained by the region of interest. This information is obtained either through the identification of known features (for example, cargo container parts, vehicle tires, windshield, etc.), HS codes, or other

ways of identifying an object. For example, a machine learning algorithm may be configured to identify tires, ceramic tiles or engine blocks inside the cargo container as these are made of known materials.

(32) FIG. 3 is a flowchart of a plurality of exemplary steps of a method **300** of processing X-ray scan images of an object under inspection for materials classification and on-the-fly automatic adaptive adjustment of calibration data, in accordance with some embodiments of the present specification. In embodiments, the method **300** comprises a plurality of programmatic instructions or code stored in a non-transient memory associated with a computing device that is in data communication with an X-ray inspection system configured to generate the X-ray scan images. The computing device includes a processor and random access memory, wherein the processor executes the method **300**.

(33) In some embodiments, the X-ray inspection system is configured to use interlaced dual energy pulses in order to generate the X-ray scan images of low energy (3-5 MeV) and high energy (6-9 MeV) of the cargo container. In various embodiments, the object under inspection may be a vehicle, truck, rail car or other containers for carrying cargo, passenger luggage or general belongings. In a non-limiting exemplary scenario, the steps of the method **300** are being described with reference to a cargo container scanned using a dual-energy X-ray inspection system. Alternate embodiments may use a single-energy X-ray inspection system. Further, embodiments may use a multi-view X-ray inspection system having single or dual-energy capabilities.

(34) At step **302**, as a result of an initial calibration process, the X-ray inspection system is configured to generate a first plurality of calibration data. The first plurality of calibration data is stored in the non-transient memory associated with the computing device. In the initial calibration process, a set of materials with known atomic numbers and densities such as plastic, aluminum, steel, lead, tungsten, with different thicknesses is put between the X-ray source and detectors (of the X-ray inspection system) and the response of the detectors, for the high and low energy beam components, is recorded corresponding to the first plurality of calibration data.

(35) At step **304**, the X-ray inspection system is operated to scan a cargo container, thereby generating data indicative of dual-energy X-ray scan images. At step **306**, the computing device is configured to apply a cargo materials segmentation algorithm, known to persons of ordinary skill in the art, to the X-ray scan images in order to identify and segment first areas or regions of interest (ROI) with known materials composition (in some embodiments, the first areas or regions of interest contain only one type of material having a known effective atomic number (Z_{eff})). For example, for an ISO container, the first areas or regions of interest include top and ends (front and rear) that may be unobstructed and known to be of steel. The unobstructed areas or regions may also include those portions of the container that are empty or devoid of the presence of any cargo or goods. In further examples, the first areas or regions of interest may additionally include tires, the windshield, vehicle frame around axles and fuel tank.

(36) It should be appreciated that calibration curves are non-linear. Although a single region or section of interest can be used for materials classification and on-the-fly automatic adaptive adjustment of calibration data, the greater the number of regions or sections of interest are processed, the more accurate the recalibration or adjustment of calibration data becomes.

(37) At step **308**, the computing device is configured to determine if one or more HS (Harmonized Commodity Description and Coding System) codes, associated with the cargo or goods of the cargo container and therefore the X-ray scan images, are known. HS codes are identification codes given to goods for use in international trade. The HS codes are administered by the World Customs Organization (WCO) and are internationally accepted for use by customs authorities and companies to identify goods/cargo.

(38) If the one or more HS codes are known, then at step **310**, the computing device is configured to further segment the cargo or goods regions of the X-ray scan images into one or more second areas or regions of interest corresponding to the one or more HS codes and subsequently the flow

moves to step **312**. At step **312**, the computing device is configured to determine first data indicative of the Zeff of the segmented first and/or second regions of interest using the first plurality of calibration data.

(39) If the one or more HS codes are not known, then the computing device is configured to determine first data indicative of the Zeff of the segmented first regions of interest using the first plurality of calibration data. The second regions of interest, which include known HS codes, are used to provide additional data points to the recalibration model. As described earlier, the segmented first regions of interest are those corresponding to known materials while the segmented second regions of interest are those corresponding to one or more HS codes. Thus, if the one or more HS codes are known, then the computing device is configured to determine first data indicative of the Zeff of the segmented first region of interest and segmented second region of interest using the first plurality of calibration data.

(40) At step **314**, the computing device is configured to compare the first data with second data indicative of reference Zeff corresponding to the known materials and/or HS codes of the segmented first and/or second regions of interest. At step **316**, the computing device is configured to determine if the first data is significantly different or deviated from the second data. In some embodiments, the first data is considered to be significantly different or deviated from the second data if the first and second data differ or deviate by at least 2 atomic number units. In some embodiments, the first data is considered to be significantly different or deviated from the second data if the first and second data differ or deviate by at least 3 atomic number units.

(41) If a significant difference or deviation is determined, as defined above, then at step **318**, the first plurality of calibration data is adjusted or modulated based on the second data in order to generate a second plurality of calibration data, the first plurality of calibration data is replaced by the second plurality of calibration data for storing in the memory, at step **320**, and subsequently the flow moves to step **322**.

(42) It should be appreciated that the adjustment or modulation of the first plurality of calibration data is non-linear. The adjustment or modulation is dependent on the number of regions or sections of interest identified or available. In some embodiments, if the identified regions or sections of interest correspond only to container components then a shift in the calibration curves (of the first plurality of calibration data) is performed to recalibrate. However, if the regions or sections of interest with HS codes are also available then each calibration curve (in the first plurality of calibration data) corresponding to the specific type of material is modified or modulated.

(43) At step **322**, the computing device is configured to calculate a Zeff value for each individual pixel in the whole X-ray scan images. In embodiments, the Zeff value for each pixel is calculated using the second plurality of calibration data if the first and second data differ significantly, as defined above. However, the Zeff value for each pixel is calculated using the first plurality of calibration data if the first and second data do not differ significantly.

(44) At step **324**, the computing device is configured to apply materials segmentation algorithm to the X-ray images in order to generate a plurality of segments and thereafter apply a smoothing or averaging filter or algorithm to the Zeff values (of step **322**) of multiple pixels, associated with each of the plurality of segments, in order to generate one or more smoothed Zeff-based segments, regions or clusters in the X-ray images. In various embodiments, the smoothing or averaging involves techniques such as, but not limited to, simple running averages to shape preserving smoothing filters, image-guided segmentation and K-means clustering.

(45) Thereafter, at step **326**, the computing device is configured to apply a predefined color map or code to the one or more smoothed Zeff-based regions of interest, segments or clusters. For example, a smoothed Zeff-based segment corresponding to organic materials are colored orange, metallic materials are colored in blue, inorganic materials are colored in green and high-Z materials are colored in red. Segments or regions with unreliable Zeff due to too little attenuation or too much noise remain uncolored (grayscale).

(46) Finally, at step 328, the computing device is configured to render or display color-coded X-ray images on a display.

(47) The above examples are merely illustrative of the many applications of the systems and methods of the present specification. Although only a few embodiments of the present invention have been described herein, it should be understood that the present invention might be embodied in many other specific forms without departing from the spirit or scope of the invention. Therefore, the present examples and embodiments are to be considered as illustrative and not restrictive, and the invention may be modified within the scope of the appended claims.

Claims

1. A method of performing an on-the-fly adjustment of calibration data corresponding to an X-ray inspection system that comprises an X-ray source in data communication with a computing device, the method being implemented in the computing device having one or more physical processors programmed with a plurality of program instructions that, when executed by the one or more physical processors, cause the computing device to perform the method, the method comprising: generating, by the X-ray inspection system, a first plurality of calibration data, wherein the first plurality of calibration data is stored in a memory in data communication with the computing device; generating, using the X-ray inspection system, data representative of X-ray scan image of a cargo container; segmenting data representative of the X-ray scan image into a plurality of regions of interest, wherein at least some of the plurality of regions of interest contain only one type of material having at least one of a known effective atomic number (Zeff) or a known identification code; determining, using the first plurality of calibration data, first data indicative of Zeff of the at least some of the plurality of regions of interest; comparing the first data with second data indicative of Zeff corresponding to the known Zeff or known identification code; adjusting the first plurality of calibration data to generate a second plurality of calibration data if the first data and the second data differ more than a predefined threshold; and using the second plurality of calibration data to process the data representative of the X-ray scan image to generate and display the processed X-ray scan image.

2. The method of claim 1, further comprising: calculating a Zeff value for each pixel in the X-ray scan image; applying a smoothing function to the Zeff values of multiple pixels, associated with each of the plurality of regions of interest, in order to generate smoothed Zeff-based regions of interest; applying a predefined color map to the smoothed Zeff-based regions of interest to generate a color-coded X-ray scan image; and displaying the color-coded X-ray scan image.

3. The method of claim 2, wherein the Zeff value for each pixel is calculated using the second plurality of calibration data if the first data and the second data differ more than the predefined threshold, and wherein the Zeff value for each pixel is calculated using the first plurality of calibration data if the first data and the second data do not differ more than the predefined threshold.

4. The method of claim 2, wherein the smoothing function comprises at least one a simple running average function, a shape preserving smoothing filter function, an image-guided segmentation function or K-means clustering function.

5. The method of claim 1, wherein the at least some of the plurality of regions of interest correspond to one or more empty regions of the cargo container or one or more unobstructed regions of the cargo container.

6. The method of claim 1, wherein at least one of the plurality of regions of interest corresponds to of a volume encompassing only a top surface of the cargo container, a volume encompassing only a front end of the cargo container.

7. The method of claim 1, wherein the first plurality of calibration data is generated by: placing, between a radiation source and detectors of the X-ray inspection system, a set of materials with

known atomic numbers and densities, wherein the set of materials have different thicknesses; and recording responses of the detectors for each material of the set of materials.

8. The method of claim 7, wherein the X-ray scan image represents recorded data corresponding to both low and high energies of the radiation source.

9. The method of claim 1, wherein at least one of the plurality of regions of interest corresponds to a volume encompassing only a front end of the cargo container.

10. The method of claim 1, wherein at least one of the plurality of regions of interest corresponds to a volume encompassing only a rear end of the cargo container.

11. An X-ray inspection system comprising: a radiation source; an array of detectors; a computing device in data communication with the radiation source and the array of detectors, wherein the computing device has one or more physical processors programmed with a plurality of program instructions that, when executed by the one or more physical processors, cause the computing device to: generate a first plurality of calibration data, wherein the first plurality of calibration data are stored in a memory associated with the computing device; generate, using data detected by the array of detectors, an X-ray scan image of a cargo container; segment the X-ray scan image into a plurality of regions of interest, wherein at least some of the plurality of regions of interest contain a single type of a material associated with a known atomic number or a single type of a material associated with a known identification code; determine, using the first plurality of calibration data, first data indicative of a first effective atomic number of the single type of the material associated with the known atomic number or a second effective atomic number of the single type of the material associated with the known identification code for at least some of the plurality of regions of interest; compare the first data with second data indicative of a known effective atomic number corresponding to the material associated with the known atomic number or a known effective atomic number corresponding to the material associated with the known identification code; adjust the first plurality of calibration data to generate a second plurality of calibration data if the first data and the second data differ more than a predefined amount; and replace the first plurality of calibration data by the second plurality of calibration data in the memory.

12. The X-ray inspection system of claim 9, wherein the plurality of program instructions, when executed by the one or more physical processors, further cause the computing device to: calculate an effective atomic number value for each pixel in the X-ray scan image; apply a smoothing function to the effective atomic number values of multiple pixels, associated with each the at least some of the plurality of regions of interest, in order to generate smoothed effective atomic number-based regions of interest; apply a predefined color map to the smoothed effective atomic number-based regions of interest to generate a color-coded X-ray scan image; and display the color-coded X-ray scan image.

13. The X-ray inspection system of claim 12, wherein the effective atomic number value for each pixel is calculated using the second plurality of calibration data if the first data and the second data differ more than the predefined amount, and wherein the effective atomic number value for each pixel is calculated using the first plurality of calibration data if the first data and the second data do not differ more than the predefined amount.

14. The X-ray inspection system of claim 12, wherein the smoothing function includes at least one of a simple running averages, shape preserving smoothing filter function, an image-guided segmentation function or a K-means clustering function.

15. The X-ray inspection system of claim 11, wherein at least some of the plurality of regions of interest are empty or unobstructed.

16. The X-ray inspection system of claim 9, wherein at least one of the plurality of regions of interest corresponds to a volume encompassing only a top surface of the cargo container.

17. The X-ray inspection system of claim 9, wherein the first plurality of calibration data is generated by: placing, between the radiation source and the array of detectors, set of materials with known atomic numbers and densities, wherein the set of materials have different thicknesses; and

recording a response of the array of detectors for each material in the set of materials.

18. The X-ray inspection system of claim 15, wherein the X-ray scan image comprises data representative low and high energies of the radiation source.

19. The X-ray inspection system of claim 11, wherein at least one of the plurality of regions of interest corresponds to a volume encompassing only a front end of the cargo container.

20. The X-ray inspection system of claim 11, wherein at least one of the plurality of regions of interest corresponds to a volume encompassing only a rear end of the cargo container.
